

Pharmacy Ordering System (InfoPharma Ltd) Software

Summary

This is the Requirements Specification and High-Level Design Document, a proposal to the Founder and CEO of InfoPharma Ltd, by Zaina Meraj, Project Manager at Softwave.

Authors

Zaina Meraj, Yaseen Barlas, Sukhmanvir Atwal, Jagdish Vaghela, Viktor Vasilchin, Rizwan Hussen, Leo Haghighi

Distribution

The Founder and CEO of InfoPharma Ltd, the Development Team at Softwave.

Contents

1 Preface

1.1 Purpose and scope of the document (what it is and what is NOT)

This document is a Requirements Specification and High-Level Design document for the InfoPharma Ordering System (IPOS). It describes the functional requirements of the IPOS-CA (Client Application) subsystem and presents the high-level design of the overall IPOS software, which consists of three interacting subsystems: IPOS-SA, IPOS-CA, and IPOS-PU.

The document explains what the IPOS-CA subsystem shall do and how it is expected to interact with the other IPOS subsystems. It does not include details of implementation such as user interface layouts, database schemas, or low-level program logic.

The document is provided in response to the initial statement of requirements issued by InfoPharma Ltd, the client for this project. UML models and high-level interface descriptions are used to convey the expected behaviour of the IPOS-CA subsystem and the mechanisms of interaction between the subsystems to system designers and software developers.

1.2 Intended audience and Reading Suggestions

The intended audience of this document includes representatives of InfoPharma Ltd, as well as system analysts, architects, designers, and software developers involved in the development of the IPOS software.

Some sections of the document are primarily intended for the development team. Readers without a technical background are encouraged to focus on the introductory sections and overview descriptions.

1.3 History of the document

This is a new document, v. 1.0.

2 Introduction

The InfoPharma Ordering System (IPOS) is intended to support the distribution and retail of pharmaceutical products by automating key business processes for InfoPharma Ltd and its pharmacy customers.

IPOS aims to improve the efficiency, accuracy, and reliability of current business operations by providing computer-based support for merchant ordering, customer account management, local stock control, sales processing, and reporting. The system is designed to reduce reliance

on manual processes and to minimise errors related to ordering, stock shortages, and customer debt handling.

The IPOS software is structured around three interconnected subsystems. IPOS-SA supports InfoPharma Ltd's internal operations, including catalogue management, merchant accounts, order processing, and invoicing. IPOS-CA is a client application used by pharmacy shops to manage local retail operations such as placing orders, managing customers, handling payments, and maintaining local stock. IPOS-PU is an online platform that enables pharmacy merchants to sell pharmaceutical products directly to members of the public.

This document focuses specifically on IPOS-CA, the subsystem responsible for supporting day-to-day operations within pharmacy shops. IPOS-CA interacts with the other subsystems as required to fulfil its functionality, but those subsystems are specified in separate documents.

3 Glossary

Actor	A human user or an external system that interacts with IPOS, such as a Pharmacist, Manager, or another IPOS subsystem.
Customer	An individual purchasing pharmaceutical products from a pharmacy using IPOS-CA. Customers may be account holders or occasional customers.
InfoPharma Ltd	The pharmaceutical distribution company for which the IPOS system is being developed.
IPOS	InfoPharma Ordering System.
IPOS-CA	Client Application subsystem of IPOS, used by pharmacy shops to manage orders, customers, stock, sales, and reports.
IPOS-SA	Server Application subsystem of IPOS, operated by InfoPharma Ltd, responsible for catalogue management, merchant accounts, and order processing.
IPOS-PU	Public-facing subsystem of IPOS that enables online sales of pharmaceutical products.

Subsystem	A major functional component of IPOS that provides a distinct set of services and interacts with other subsystems via defined interfaces.
UML	Unified Modelling Language, used to model system requirements and high-level design.
Use Case	A description of a basic interaction between an actor and the system to achieve a specific goal.

4 User Requirements

This section is left intentionally blank

5 System Requirements

5.1 Purpose and scope of the system

This section provides a more detailed specification of required functionality (known as system requirements) for IPOS-CA.

The intended audience is the system architects, designers, and software developers for the implementation of IPOS-CA. Since we use UML, which requires a degree of specific expert knowledge, we do not expect this section to be immediately useful to the InfoPharma staff but would like to encourage them to review and comment if possible

5.2 System Models

5.2.1 UML use-case models

5.2.1.1 Use case diagram

The use case diagram for IPOS-CA subsystem is shown below in Figure 1. In this diagram the other subsystems of IPOS-CA are treated as “external systems”, with which IPOS-CA may interact.

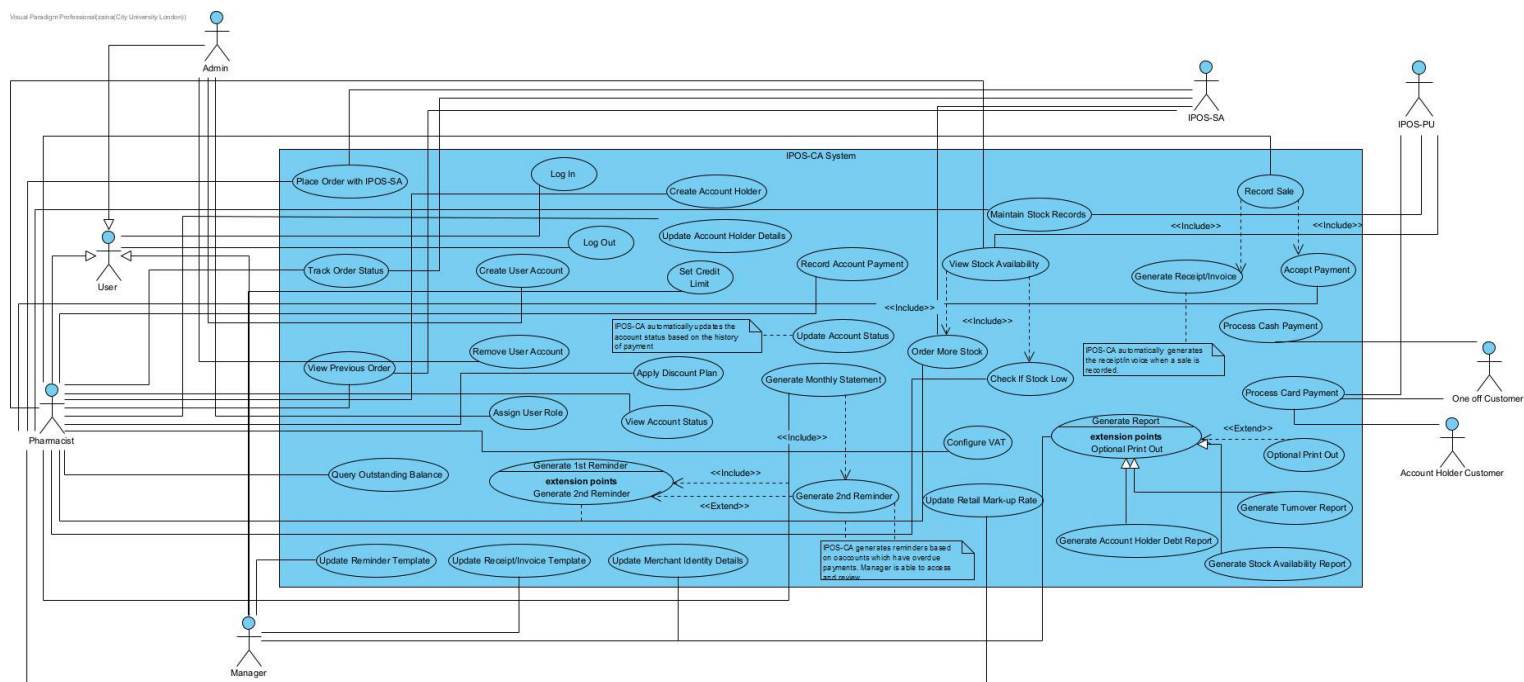


Figure 1. Use case diagram of IPOS-CA. The use cases cover the required functionality of the IPOS-CA subsystem (see the initial statement of requirements). The actors represent the external entities (users or 3rd party systems) that interact with IPOS-CA. The other subsystems of IPOS, IPOS-SA, and IPOS-PU are also modeled as actors.

5.2.1.1 Use case specifications (simplified)

ID: CA-01	Use Case: Log In
Brief Description: Allows the pharmacist to authenticate themselves to gain access into the IPOS-CA subsystem. The system verifies the log in credentials of the staff before allowing access into the subsystem.	
Primary Actors: User	
Secondary Actors: None	

Preconditions: 1. System must be functional. 2. User must have an existing account with valid log in credentials.	
Postconditions: 1. User is successfully logged into the system.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InvalidCredentials 2. LockedAccount	

ID: CA-02	Use Case: Log Out
Brief Description: Allows a user who is currently logged into the system to successfully logout of the system to prevent any unauthorised access into the pharmacy system.	
Primary Actors: User	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. User is logged into the system using valid credentials.	
Postconditions: 1. The user must not have access in the pharmacy system at all after logging out.	

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. SessionTimedOut

ID: CA-03**Use Case:** Create User Account**Brief Description:**

Allows for a customer account to be created in the pharmacy system, which takes the customer's first name, surname, DOB, email address, home address and phone number. This is done to store customer details on the pharmacy system for future transactions.

Primary Actors:

Admin

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. Admin is logged into the system.
3. Inputs of valid first name, surname, DOB, email address, home address and phone number into required fields.

Postconditions:

1. Customer's data is recorded and stored onto the pharmacy system successfully.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidInput (e.g., user enters '#' within the surname field). 2. DuplicateCustomer

ID: CA-04	Use Case: Remove User Account
Brief Description: Removes the existing user account, preventing access into the IPOS-CA system.	
Primary Actors: Admin	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. User account exists in the IPOS-CA system. 3. Admin is aware of which account to be removed.	
Postconditions: 1. Account has been successfully removed from the system.	
Extension Use Case: None	

Inclusion Use Case: None
Alternative Flows: 1. UserNotFound

ID: CA-05	Use Case: Update Account Holder Details
Brief Description: Updates the existing details which are stored of an existing account holder. This is done to ensure account holder details remain accurate.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. There is an existing account holder. 3. The pharmacist is logged into the system successfully using their credentials. 4. The pharmacist aware of the details which need to be updated.	
Postconditions: 1. Account holder details are updated successfully.	
Extension Use Case:	

None

Inclusion Use Case:

None

Alternative Flows:

1. AccountHolderNotExisting
2. InvalidDetailsToUpdate

ID: CA-06

Use Case: View Account Status

Brief Description:

Displays the status of the account holder, which includes the balance and checking statements.

Primary Actors:

Pharmacist

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. There is an existing account holder.
3. The pharmacist will access the account holder status.

Postconditions:

1. Account status is displayed to the user

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. AccountHolderNotExisting
2. AccountHolderSuspended

ID: CA-07**Use Case:** Generate Monthly Statements**Brief Description:**

A monthly statement will be generated which contains all transactions which have been made within that particular month, helping to support the customer in reviewing their past transactions.

Primary Actors:

Pharmacist

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. There is an existing account holder.
3. Monthly transactions are present within the system.

Postconditions:

1. Monthly statements are generated successfully.

Extension Use Case:

None

Inclusion Use Case:

• Generate 1st Reminder • Generate 2nd Reminder
Alternative Flows: 1. NoTransactionsMade

ID: CA-08	Use Case: Assign User Role
Brief Description: A role will be assigned to the user account, which demonstrates how much of the system can be accessed within the pharmacy's subsystem.	
Primary Actors: Admin	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. A user account exists on the system. 3. Admin has the correct login credentials to access the system. 4. Admin is aware of the appropriate role to assign to each user.	
Postconditions: 1. Admin has successfully assigned a role to a user account.	
Extension Use Case: None	
Inclusion Use Case: None	

Alternative Flows:

1. InvalidRoleEntered
2. NoUserPresent

ID: CA-09**Use Case:** Update Account Status**Brief Description:**

Updates the status of the account holder, such as the account being active or suspended.

Primary Actors:

IPOS-CA

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. Account holder exists in the system.
3. The subsystem is aware of which account to update the status on.

Postconditions:

1. Account holder's status is updated.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. AccountNotFound

2. InvalidStatusProvided

ID: CA-10	Use Case: Record Account Payment
Brief Description: Any payments made by the account holder is recorded, updating the outstanding balance.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. Account holder has an outstanding balance which has been already paid.	
Postconditions: 1. Payment is recorded. 2. Outstanding balance is updated.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. PaymentDeclined	

ID: CA-11	Use Case: Generate 1 st Reminder
Brief Description: A first reminder is generated for accounts which have overdue payments.	
Primary Actors: IPOS-CA	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. Account holder exists in the system. 3. Account holder has a payment which is overdue.	
Postconditions: 1. Reminder will be generated.	
Extension Use Case: 1. Generate 2nd Reminder	
Inclusion Use Case: None	
Alternative Flows: 1. FailToSendReminder	

ID: CA-12	Use Case: Generate 2 nd Reminder
Brief Description: A second reminder will be generated if the payment is still overdue after the first reminder.	
Primary Actors: IPOS-CA	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. First reminder is already sent to the account holder. 3. Second reminder is sent to the account holder.	
Postconditions: 1. Second reminder has been generated and sent to the account holder.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. FailedToSendReminder	

ID: CA-13	Use Case: Record Sale
Brief Description: A record is made of every sale transaction which is completed within the pharmacy system.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. A sale is completed on the pharmacy system.	
Postconditions: 1. Sale is recorded.	
Extension Use Case: None	
Inclusion Use Case: 1. Generate Receipt/Invoice 2. Accept Payment	
Alternative Flows: 1. SaleRecordingFailed	

ID: CA-14	Use Case: Accept Payment
Brief Description: A payment will be accepted for a sale or any account transactions which have been made.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. A payment will be required for any pharmaceutical product which is being purchased.	
Postconditions: 1. Payment will be accepted.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. PaymentFailed	

ID: CA-15	Use Case: Process Cash Payment
Brief Description: Processes payments which are made using cash.	
Primary Actors: One off Customer	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. Customer picks a pharmaceutical product they'd like to purchase 3. Method of payment selected by the customer is cash.	
Postconditions: 1. Cash payment is processed.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InsufficientCash	

ID: CA-16	Use Case: Process Card Payment
Brief Description: Processes payments which are made using a card.	
Primary Actors: 1. Customer (One off Customer or Account Holder Customer)	
Secondary Actors: 1. IPOS-PU	
Preconditions: 1. System is fully operational. 2. Customer chooses a pharmaceutical product they'd like to purchase. 3. Method of payment chosen by the customer is by card.	
Postconditions: 1. Card payment is processed.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InsufficientFundsInCard	

ID: CA-17	Use Case: Generate Receipt/Invoice
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Brief Description:	
A receipt/invoice is generated when a successful transaction is completed.	
Primary Actors:	
IPOS-CA	
Secondary Actors:	
None	
Preconditions:	
1. System is fully operational. 2. Payment is successfully completed for the pharmaceutical product the customer wanted to purchase.	
Postconditions:	
1. Receipt/Invoice is generated by the subsystem (IPOS-CA).	
Extension Use Case:	
None	
Inclusion Use Case:	
None	
Alternative Flows:	
1. PrintError 2. PaymentFailed	

ID: CA-18	Use Case: Maintain Stock Records
Brief Description:	

Provides accurate records for the level of stock of pharmaceutical products.	
Primary Actors:	
Pharmacist	
Secondary Actors:	
IPOS-PU	
Preconditions:	
1. Systems are fully functional. 2. There is existing inventory of pharmaceutical products. 3. Pharmacist and IPOS-PU subsystem is aware of which product's stock levels need updating.	
Postconditions:	
1. Stock records are updated.	
Extension Use Case:	
None	
Inclusion Use Case:	
None	
Alternative Flows:	
1. NoStock 2. InventoryUpdateFailed	

ID: CA-19	Use Case: View Stock Availability
Brief Description: Allows the merchant to view current stock levels for all products, including quantities available and basic product details.	

Primary Actors:

Pharmacist

Secondary Actors:

IPOS-PU

Preconditions:

1. Systems are fully functional.
2. Pharmacist is authenticated.
3. Product and stock data exists within the system.

Postconditions:

1. Current stock availability is displayed to Pharmacist

Extension Use Case:

None

Inclusion Use Case:

1. Order More Stock
2. Check If Stock Low

Alternative Flows:

None

ID: CA-20

Use Case: Check If Stock low

Brief Description:

Automatically checks stock level against the predefined thresholds and identifies products with low stock

Primary Actors: Pharmacist
Secondary Actors: None
Preconditions: 1. System is fully functional. 2. Stock availability data exists. 3. Low-stock thresholds are defined.
Postconditions: 1. Low-stock items are highlights/flagged
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. DataNotRetrieved

ID: CA-21	Use Case: Configure VAT
Brief Description: Allows the Pharmacist to configure VAT rates that are applied to products and transactions.	
Primary Actors: Pharmacist	

Secondary Actors: None
Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of what VAT to configure.
Postconditions: 1. New VAT configuration is saved and applied to future calculations.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidVATEntered

ID: CA-22	Use Case: Update Retail Mark-up Rate
Brief Description: Allows the Pharmacist to configure markup rates that are applied to products and transactions.	
Primary Actors: Pharmacist	
Secondary Actors:	

None
Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of the mark-up rates to configure on products and transactions.
Postconditions: 1. New markup rates are saved and applied to future calculations.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidRateEntered

ID: CA-23	Use Case: Order More Stock
Brief Description: Allows Pharmacist to place an order for additional stock	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-SA	
Preconditions:	

1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of the product which needs reordering.
Postconditions: 1. Order is submitted to the IPOS-SA subsystem.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. OrderSubmissionFails

ID: CA-24	Use Case: Generate Report
Brief Description: Generates a report, is a generalized class in which the specialised will use.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions:	

1. System is fully functional. 2. Manager is logged into the system.
Postconditions: 1. The requested report is generated for the specified period. 2. The report is displayed on screen.
Extension Use Case: 1. Optional Printout
Inclusion Use Case: None
Alternative Flows: 1. NoDataAvailableForPeriod

ID: CA-25	Use Case: Optional Printout
Brief Description: After a report is generated, it gives the manager the option to create a printout.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. A report has already been generated and displayed.	

Postconditions:

2. Report is generated successfully.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. Printer used to create printout fails.

ID: CA-26

Use Case: Generate Turnover Report

Brief Description:

Generates a report showing sales turnover for a selected period.

Primary Actors:

Manager

Secondary Actors:

None

Preconditions:

Inherits preconditions from Generate Report.

Postconditions:

1. Turnover report is generated successfully.
Extension Use Case: Inherits extension Optional Printout
Inclusion Use Case: None
Specialisation of: Generate Report
Alternative Flows: 2. NoDataAvailableForPeriod

ID: CA-27	Use Case: Generate Stock Availability Report
Brief Description: Generates a report of current stock levels.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: Inherits preconditions from Generate Report.	
Postconditions: 1. Stock availability report is generated successfully.	

Extension Use Case: Inherits extension Optional Printout
Inclusion Use Case: None
Specialisation of: Generate Report
Alternative Flows: 1. StockLevelError

ID: CA-28	Use Case: Generate Account Holder Debt Report
Brief Description: Generates a report of outstanding account holder debts.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: Inherits preconditions from Generate Report.	
Postconditions: 1. Account debt report is generated successfully.	
Extension Use Case:	

Inherits extension Optional Printout
Inclusion Use Case: None
Specialisation of: Generate Report
Alternative Flows: 1. NoAccountHolderWithDebt

ID: CA-29	Use Case: Place Order with IPOS-SA
Brief Description: Allows an order to be placed with the supplier's subsystem, which is IPOS-SA.	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-SA	
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of which order of products need to be placed.	
Postconditions: 1. Order is registered and placed with IPOS-SA.	
Extension Use Case:	

None
Inclusion Use Case: None
Alternative Flows: 1. SupplierSystemUnavailable

ID: CA-30	Use Case: Track Order Status
Brief Description: Allows tracking the status of an order placed with IPOS-SA.	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-SA	
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Order has already been placed before by the pharmacist.	
Postconditions: 1. Order status is displayed.	
Extension Use Case: None	

Inclusion Use Case: None
Alternative Flows: 1. OrderNotFound

ID: CA-31	Use Case: View Previous Order
Brief Description: Allows viewing of previously placed orders.	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-SA	
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Previous orders exist, which were made by the pharmacist.	
Postconditions: 1. List of previous orders is displayed.	
Extension Use Case: None	
Inclusion Use Case:	

None

Alternative Flows:

1. NoPreviousOrdersMade

ID: CA-32

Use Case: Query Outstanding Balance

Brief Description:

Allows querying an account holder's outstanding balance.

Primary Actors:

Pharmacist

Secondary Actors:

None

Preconditions:

1. System is fully functional.
2. Pharmacist is logged into the system using their credentials.
3. Account holder exists who still yet has an outstanding balance (which the pharmacist can see).

Postconditions:

1. Outstanding balance is displayed to the pharmacist.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. AccountHolderNotFound

ID: CA-33

Use Case: Set Credit Limit

Brief Description:

Allows the updating/setting of a credit limit for an account holder.

Primary Actors:

Manager

Secondary Actors:

None

Preconditions:

1. System is fully functional.
2. Manager is logged into the system using their credentials.
3. Account holder exists.
4. Manager is aware of which account holder requires a credit limit set.

Postconditions:

1. Credit limit is updated/set.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. InvalidCreditLimitEntered

ID: CA-34**Use Case:** Apply Discount Plan**Brief Description:**

Applies a discount plan to an account holder.

Primary Actors:

Pharmacist

Secondary Actors:

None

Preconditions:

1. System is fully functional.
2. Pharmacist is logged into the system using their credentials.
3. Account holder exists.
4. Pharmacist is aware which account holder requires a discount plan to be applied.

Postconditions:

1. Discount plan is applied to the account holder.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. InvalidDiscountPlanSelected

ID: CA-35**Use Case:** Update Reminder Template**Brief Description:**

Allows the updating of the template used for payment reminders.

Primary Actors:

Manager

Secondary Actors:

None

Preconditions:

1. System is fully functional.
2. Manager is logged into the system using their credentials.
3. Manager is aware which payment reminder template needs to be updated.

Postconditions:

1. Reminder template is updated.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. InvalidPaymentTemplateFormat

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ID: CA-36	Use Case: Update Receipt/Invoice Template
Brief Description: Allows the updating of the receipt template.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Manager is logged into the system using their credentials. 3. Manager is aware which receipt template needs to be updated.	
Postconditions: 1. Receipt template is updated.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InvalidReceiptTemplateFormat	

ID: CA-37	Use Case: Update Merchant Identity Details
Brief Description: Allows updating the pharmacy's identity details used on receipts and documents.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Manager is logged into the system using their credentials. 3. Manager is aware which merchant's identity details need to be updated.	
Postconditions: 1. Merchant identity details are updated.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InvalidDetailsProvided	

5.2.1.3 Prioritised List of use cases

The list below prioritises the use cases included in the use case model according to their importance for IPOS-CA. The development team will use the assigned priorities to decide on the order of implementing the functionality specified in the use cases according to the resources available: the essential use cases will be implemented first, then the use cases ranked as important will follow. The use cases ranked as good-to-have will be implemented last, provided the resources (manpower and time) are available to complete their implementation.

Use case ID	Use case name	UC priority	Justification
CA-01	Log In	Essential	Required to restrict access to authorised pharmacy staff and enable role-based operation of the system.
CA-02	Log Out	Good to have	Improves security, but the system could still function with session timeout/auto-logout as a workaround.
CA-03	Create User Account	Essential	Customer/account-holder records must exist to support credit, balance tracking, and repeat transactions.
CA-04	Remove User Account	Important	Needed for correct administration and preventing future use, but daily operation can continue without it short-term.
CA-05	Update Account Holder Details	Important	Keeps customer data accurate for operations, but limited inaccuracies wouldn't stop core selling/ordering.
CA-06	View Account Status	Essential	Staff must be able to view balance/status to decide whether purchases on account are allowed.

CA-07	Generate Monthly Statements	Important	Supports the monthly debt process, manual statements could be produced temporarily if needed.
CA-08	Assign User Role	Essential	Role-based access is required to separate pharmacist/admin/manager capabilities and protect functions.
CA-09	Update Account Status	Important	Needed to manage account conditions (active/suspended), but some status changes could be handled later.
CA-10	Record Account Payment	Essential	Payments must be recorded to update balances and allow customers to return to normal status.
CA-11	Generate 1st Reminder	Important	Improves debt recovery and supports credit control, manual reminders could temporarily substitute.
CA-12	Generate 2nd Reminder	Good to have	Adds escalation but the system remains usable with only one reminder or manual escalation.
CA-13	Record Sale	Essential	Core retail function, without recorded sales, stock and financial tracking becomes unreliable.
CA-14	Accept Payment	Essential	Sales and debt clearance require payment acceptance, as without it the system cannot complete transactions.
CA-15	Process Cash Payment	Important	Many pharmacies accept cash. However, the system could still operate if card-only was used temporarily.
CA-16	Process Card Payment	Essential	Card payments are a primary method and required for account holders' payments, critical for normal operation.

CA-17	Generate Receipt/Invoice	Important	Improves customer proof/accounting, but transactions could still be completed without printing receipts initially.
CA-18	Maintain Stock Records	Essential	Stock records are required to ensure items can be sold and re-ordered accurately.
CA-19	View Stock Availability	Essential	Staff must see current stock to sell correctly and avoid selling unavailable items.
CA-20	Check If Stock Low	Important	Low stock alerts reduce shortages, but staff could monitor stock manually in the short term.
CA-21	Configure VAT	Important	VAT affects pricing and receipts. System can operate with a default VAT until configuration is added.
CA-22	Update Retail Markup Rate	Important	Supports correct pricing strategy, system can operate using a default markup temporarily.
CA-23	Order More Stock	Essential	Replenishing stock is critical to the pharmacy, without reordering, the system fails a core business need.
CA-24	Generate Report	Essential	A generalized class that is used for the specialized classes for generating reports.
CA-25	Optional Printout	Good to have	After a report is generated, the manager gets the choice to print out said report.
CA-26	Generate Turnover Report	Good to have	Useful for management insight but not required for daily operations (selling/stock/order).

CA-27	Generate Stock Availability Report	Good to have	Convenient summary for managers, core stock viewing already exists via other use cases.
CA-28	Generate Account Debt Report	Important	Supports managing overdue accounts at scale, could be approximated manually from account status/balances.
CA-29	Place Order with IPOS-SA	Essential	Required to interact with supplier system for ordering, without it the pharmacy cannot replenish from InfoPharma.
CA-30	Track Order Status	Important	Helps manage deliveries and planning, but orders could still be placed without tracking initially.
CA-31	View Previous Orders	Important	Useful for reordering and auditing, not strictly required to place new orders in the short term.
CA-32	Query Outstanding Balance	Essential	Staff must know customer debt to enforce credit limits and payment rules before allowing purchases.
CA-33	Set Credit Limit	Essential	Credit control is central to account-holder functionality, debt management cannot be enforced without it present.
CA-34	Apply Discount Plan	Important	Discounts are a business feature; system can operate without discounts initially, but value is reduced.
CA-35	Update Reminder Template	Good to have	Template editing is convenience/customisation. Reminders can still be generated using default wording.
CA-36	Update Receipt Template	Good to have	Custom formatting is not essential. Default receipt format is sufficient for operation.

CA-37	Update Merchant Identity Details	Good to have	Cosmetic/branding info is not required for core functionality, can be added later.
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6 Anticipated Evolution of System Requirements

We expect that IPOS-CA will be developed and deployed quickly. The operational experience from merchants and pharmacy staff will inform future decisions about necessary changes. The envisaged likely changes to various parts of IPOS-CA are summarised below.

6.1 Payment Integration

Currently, account holder payments are recorded manually by the pharmacist. In the future, IPOS-CA may integrate directly with a card payment processor to automate the recording of payments and reduce the risk of human error.

6.2 Email Communication

Currently, reminders and statements generated by IPOS-CA are produced as printable documents intended to be sent by recorded post. In the future, IPOS-CA may use the IPOS-PU-COMMS email service directly to send reminders and statements electronically to account holders, reducing postage costs and improving efficiency.

6.3 Discount Plan Expansion

Currently, two discount plan types are supported—fixed and flexible. In the future, additional discount plan types may be introduced to accommodate more complex promotional arrangements as merchant needs evolve.

6.3 Membership Cards

In the production version, account holders will be issued physical membership cards to confirm their identity at the point of sale. In the prototype, account holders confirm membership by providing their account number verbally. Future versions of IPOS-CA will need to accommodate card scanning functionality.

6.4 VAT Rate Configurability

Currently a single VAT rate applies to all goods. As UK pharmaceutical VAT regulations are subject to change, future versions may require more granular VAT configuration at the individual product level.

7 IPOS-CA High -Level Design

This section defined the high-level design of the entire IPOS-CA software, which includes its architecture—the subsystems that it is made up of and the communication interfaces between the subsystems. The subsystems envisaged are:

7.1 IPOS-SA

A brief paragraph describing what IPOS-SA does—the server application administered by the InfoPharma IT department—maintains an electronic catalog and handles merchant accounts, orders, payments, and reports.

7.2 IPOS-CA

A brief paragraph describing what IPOS-CA does—the client desktop application run by the merchant—handles local stock, customer accounts, sales, reminders, and reports.

7.3 IPOS-PU A brief paragraph describing what IPOS-PU does—the online portal for public sales, membership, promotion campaigns, and communication services.

7.4 Software Architecture

The architecture of IPOS-CA as a UML component diagram, developed with Visual Paradigm, Professional Edition, is shown on Figure 2.

Figure 2. IPOS architecture as a UML component diagram. Each subsystem is modeled as a component with the stereotype <<subsystem>>. The communication between the subsystems is achieved via a set of assembly connectors (i.e., a pair of "provided" and "required" interfaces referring to the API offered by the respective subsystems). LoginAPI, OrderAPI, and CatalogueAPI are provided by IPOS-SA; InventoryAPI is provided by IPOS-CA, and EmailAPI, PaymentAPI, and CommercialApplicationAPI are provided by IPOS-PU. The APIs are used to access the functionality offered by one of the subsystems by the other two subsystems.

7.4. Design diagram of subsystem IPOS-CA

Next, we elaborate the high-level design of IPOS-CA and provide details on the interfaces that this subsystem needs, and on the implementation of the subsystem's provided interface. The class diagram of IPOS-CA subsystem is shown in Figure 3.

7.5 Source Code (Java) for the Interfaces of IPOS-CA

Now that we have defined all interfaces and the implementation class `CAInventoryAPI_Implementation`, source code (e.g., Java source code) for the interfaces and the implementation classes shown in Figure 3. This can be done with Visual Paradigm. The complete Java code of the interfaces and a skeleton of the implementation class generated by Visual Paradigm is provided below:

7.6 Testing the Communications Between Subsystems

Now that we have defined the interfaces necessary for communication between the subsystems of IPOS, and generated Java code for all of them, we develop tests for inter-subsystem communication, i.e. of the respective implementation classes. Here we sketch tests for the methods of the implementation class `CAInventoryAPI_Implementation` without committing to a particular testing platform to run the tests (e.g. JUnit).

8 Appendices

8.1 Initial Statement of Requirements by the Client

The InfoPharma Ordering System (IPOS)

Your company has been invited to develop an IT system to assist with the distribution of pharmaceutical goods for InfoPharma Ltd, a major pharmaceutical distributor. InfoPharma intend to optimise their interaction with their clients, retail merchants (individuals running own chemist shops and large retailers with chains of stores, e.g. Boots and Superdrugs), who sell drugs and other pharmaceutical goods to members of the public.

IPOS includes three subsystems:

- i) A server application (IPOS-SA), which in the fully functional production version will be administered by the IT department of InfoPharma Ltd.
- ii) A client application (IPOS-CA), a copy of which will be run by the merchant on a desktop PC in the merchant's pharmacy shop.
- iii) An online platform (IPOS-PU) to be used by a merchant for direct sales of pharmaceutical goods to the public.

The main functionality required from IPOS-CA is summarised below:

IPOS-CA-ORD: This package allows the merchant to place an order with IPOS-SA, track its progress through IPOS-SA (shipping, delivery), view the previous orders and query own outstanding balance. A merchant is expected to access IPOS-SA-ORD by using a username and password created for them by the administrator of IPOS-SA and sent to them using a reliable delivery means, e.g. a recorded post. Since IPOS-SA holds the merchant's username and password, IPOS-CA needs to access them via IPOS-SA to allow the merchant to login.

IPOS-CA-USER: This package should implement a light touch authentication (a username/password) with role-based access to the functionality of IPOS-CA. As a minimum the following roles of users must be implemented: i) Pharmacist: the user who can use all packages except IPOS-CA-USER; ii) Admin: will be able to deal with user accounts for IPOS-CA (create a new user account with a specific role and remove an account); iii) Manager: this role should be able to generate the reports envisaged in IPOS-CA-RPT.

IPOS-CA-CUST: Maintain a database of account holders (consumers regularly buying pharmaceutical and beauty products from the merchant). Account holders are given credit up to a predefined limit, can be given a discount (the same two types of discount plans as for IPOS-SA must be implemented—fixed and flexible) on the purchases, and are allowed to pay by the end of the calendar month. Between the 5th and 15th of every month, IPOS-CA should allow the merchant to generate statements for all those account holders who have not cleared their balances at the end of the previous calendar month. In the statement, the account holders will be asked to pay in full by the 15th of the current month. If account holders are late with their payments, the system should allow the merchant to produce two reminders about late payment. IPOS-CA should maintain automatically (i.e. without human intervention) the status of an account, which changes from 'normal' to 'suspended' and finally to 'in default' based on payment history.

IPOS-CA-Templates: This package allows Managers to change the templates used for reminders and receipts/invoices and details related to the merchant identity (e.g., pharmacy name, logo, address, and email).

IPOS-CA-Stock: This package should maintain a database with the local stock availability and provide a warning of low stock to help the merchant order new stock from InfoPharma in due time. The structure of the stock records should be similar to the catalogue structure in IPOS-SA except that an extra field will be added to carry the retail markup rate to be applied to the bulk cost of the item as received from InfoPharma. VAT will also be added at a rate common for all goods. This parameter can be set to 0% but may change and therefore must be configurable.

IPOS-CA-Sales: This package allows for sales to customers (account holders and occasional customers) to be recorded in IPOS-CA including payments. A consumer can make payments using cash and credit/debit cards. The system should store the form of payment and the amount received. In the case of card payments, the card type, the first four and the last four digits of the card number, and the expiry date should also be stored. Account holders are only allowed to use cards (credit or debit) for their payments. Occasional customers (i.e., those

who do not have an account with the merchant) can use both forms of payment, but their payments should be made in full at the time of purchasing pharmaceutical goods. A successful sale leads to producing a receipt/invoice.

IPOS-CA-RPT: This package allows the Manager to generate the following reports:

- Turnover within a given report period (amount of sales, value of new orders placed with InfoPharma).
- Stock availability report: items, availability, and value of stock (and VAT).
- Change of aggregated debt by account holders for a given period: aggregated debt at the beginning of the period, payments received from all debtors within the period, and new aggregated debt from all account holders at the end of the period.

InfoPharma Ltd. may decide to offer your company the development of the production version of IPOS and a maintenance contract for it, if you do a good job with the prototype of IPOS.

In the prototype of IPOS the three subsystems should be implemented as separate desktop applications able to interact with one another as necessary to meet the requirements. Online sales via IPOS-PU should be linked to the inventory of the merchant — successful online purchases should be deducted from the stock of goods available at the merchant's shop. Likewise, orders from IPOS-SA delivered to a merchant should be easy to record in IPOS-CA and increase the merchant's stock. Finally, promotion campaigns run via IPOS-PU may affect the cost of selling goods online at reduced price, but not the cost of the same goods if sold in the pharmacy shops.

InfoPharma Ltd. specifically requested that the new system must improve upon the current catalogue-based operation. The system should be intuitive and easy to use for pharmacy staff with varying levels of technical expertise.

8.2 Hardware (minimal and optimal configuration)

8.2.1 Minimal hardware configuration

The minimal hardware configuration recommended for IPOS-CA development consists of a desktop PC, 4 GB of RAM and at least 100 GB of disk space for storing the software programs and data, possibly running a database server such as MySQL or similar. A standard desktop or laptop machine with a stable internet connection is required to ensure connectivity with IPOS-SA and IPOS-PU during operation.

8.2.2 Optimal hardware configuration

The optimal hardware configuration envisaged for development should consist of a desktop PC, 8 GB of RAM and at least 250 GB of disk space for storing the software programs and data, possibly running a database server such as PostgreSQL or better. A stable broadband

internet connection is recommended to ensure smooth inter-subsystem communication between IPOS-CA, IPOS-SA and IPOS-PU during both development and live deployment.

9 Index

Not provided in this release of the document